

示例场景验证检查清单

用于验证剩余示例场景的检查清单。

注： `MptkIntegrationSampleScene.unity` 另外需要 `FEATURE_USE_MPTK`。

通用前置准备

各场景共同确认的项目：

- ☐ 打开场景并进入 **Play 模式**（控制台无错误）
- ☐ 打开 `Window > MIDI > Monitor`，按需确认 IN/OUT
- ☐ 启用相应的 **Scripting Define Symbols**（见下表）
- ☐ IMGUI 面板显示，并对按钮操作有响应

Scripting Define Symbols 一览

符号	目标场景
FEATURE_USE_TIMELINE + com.unity.timeline	MidiTimelineIntegrationSampleScene
FEATURE_USE_VISUALSCRIPTING + com.unity.visualscripting	MidiVisualScriptingIntegrationSampleScene
FEATURE_MIDI_NETWORK	MidiNetworkJamSampleScene
FEATURE_INPUT_SYSTEM + com.unity.inputsystem	InputSystemBridgeSampleScene

场景一览

#	类别	场景	路径
1	游戏玩法基础	MidiGameplaySampleScene	Assets/MIDI/Samples/Gameplay/Scenes/MidiGameplaySampleScene.unity
2	Unity 集成	MidiAnimatorIntegrationSampleScene	Assets/MIDI/Samples/Integrations/Scenes/MidiAnimatorIntegrationSampleScene.unity
3	Unity 集成	MidiTimelineIntegrationSampleScene	Assets/MIDI/Samples/Integrations/Scenes/MidiTimelineIntegrationSampleScene.unity
4	Unity 集成	MidiVisualScriptingIntegrationSampleScene	Assets/MIDI/Samples/Integrations/Scenes/MidiVisualScriptingIntegrationSampleScene.unity
5	游戏玩法	ChordPuzzleSampleScene	Assets/MIDI/Samples/Gameplay/Scenes/ChordPuzzleSampleScene.unity
6	游戏玩法	ChordScaleSampleScene	Assets/MIDI/Samples/Gameplay/Scenes/ChordScaleSampleScene.unity
7	游戏玩法	MidiClockSyncSampleScene	Assets/MIDI/Samples/Gameplay/Scenes/MidiClockSyncSampleScene.unity
8	游戏玩法	ScalePracticeSampleScene	Assets/MIDI/Samples/Gameplay/Scenes/ScalePracticeSampleScene.unity
9	Input System	InputSystemBridgeSampleScene	Assets/MIDI/Samples/Integrations/InputSystem/Scenes/InputSystemBridgeSampleScene.unity
10	网络	MidiNetworkJamSampleScene	Assets/MIDI/Samples/Integrations/Networking/Scenes/MidiNetworkJamSampleScene.unity
11	Foundation	FoundationSampleScene	Assets/MIDI/Samples/Foundation/Scenes/FoundationSampleScene.unity

推荐测试顺序

1. **基础**：MidiGameplaySampleScene
2. **Unity 集成**：Animator → Timeline → Visual Scripting
3. **横切基础**：ClockSync → Chord/Scale → InputSystem
4. **网络 / Foundation**：MidiNetworkJamSampleScene → FoundationSampleScene

分场景检查清单

1. MidiGameplaySampleScene

路径：Assets/MIDI/Samples/Gameplay/Scenes/MidiGameplaySampleScene.unity

目的：DeviceFilter → ChannelFilter → Router / NoteTracker 管线

- ☐ Play 开始后, IMGUI 显示管线状态
 - ☐ **NoteOn C4 (ch0)** → Router 的 Launch 绑定触发 (日志 / 计数器更新)
 - ☐ **CC64 = 127 (ch0)** → Toggle 绑定触发
 - ☐ **NoteOn C4 (ch1)** → 被 ChannelFilter 拦截 (不到达 Router / Tracker)
 - ☐ **Add D4 + E4 (ch0)** → 3 音符和弦检测消息
 - ☐ SmfPlayer 播放 / MidiRecorder 录制部分可工作
 - ☐ 连接实体 MIDI 控制器时, ch0 的 C4 / CC64 行为相同
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2. MidiAnimatorIntegrationSampleScene

路径：Assets/MIDI/Samples/Integrations/Scenes/MidiAnimatorIntegrationSampleScene.unity

目的：MIDI → Animator 参数驱动

- ☐ **CC1 = 127** → 立方体上升
 - ☐ **Note 60** → 脉冲效果 (缩放变化)
 - ☐ **CC10 / CC11** → 立方体旋转
 - ☐ 虚拟 MIDI Inject 与实体输入均可产生相同变化
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3. MidiTimelineIntegrationSampleScene

路径：Assets/MIDI/Samples/Integrations/Scenes/MidiTimelineIntegrationSampleScene.unity

前提：FEATURE_USE_TIMELINE、com.unity.timeline

- ☐ **Play Timeline** → SMF 开始播放
 - ☐ Director 时间与 SmfPlayer 时间同步
 - ☐ **Pause** → SmfPlayer 暂停
 - ☐ **Seek to 1.0 s** → SmfPlayer 跟随
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4. MidiVisualScriptingIntegrationSampleScene

路径：Assets/MIDI/Samples/Integrations/Scenes/MidiVisualScriptingIntegrationSampleScene.unity

前提：FEATURE_USE_VISUALSCRIPTING、com.unity.visualscripting

- ☐ **NoteOn C4** → 立方体颜色变化
 - ☐ **CC1 = 127** → 视觉反馈变化
 - ☐ 事件经 Event Bus / Script Graph 到达
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5. ChordPuzzleSampleScene

路径：Assets/MIDI/Samples/Gameplay/Scenes/ChordPuzzleSampleScene.unity

目的：和弦名称测验 (MidiScaleQuiz)

- ☐ 画面显示 Target chord
 - ☐ **Hold C Major / C Minor / Cmaj7** → 识别和弦出现在日志中
 - ☐ **Check Answer** → Correct / Incorrect 判定
 - ☐ 答对时 **Next Puzzle** 进入下一题
 - ☐ **Release All** 释放音符
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6. ChordScaleSampleScene

路径：Assets/MIDI/Samples/Gameplay/Scenes/ChordScaleSampleScene.unity

目的：ChordRecognition + MidiChordDetector + MidiScaleQuiz

- ☐ **Hold C Major** → Chord = C, C Major scale = OK
 - ☐ **Hold C Minor** → Chord = Cm
 - ☐ **Add Out-of-scale Note (61)** → Out of scale 显示 / 日志
 - ☐ **Evaluate Quiz Target = C** → 测验判定
 - ☐ **Release All** 重置状态
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7. MidiClockSyncSampleScene

路径：Assets/MIDI/Samples/Gameplay/Scenes/MidiClockSyncSampleScene.unity

目的：外部 MIDI Clock 同步与 BPM 估算

- ☐ **Inject Start** → IsPlaying = true
 - ☐ **Inject 24 Timing Clocks** → BPM 估算值更新, Bar/Beat 前进
 - ☐ **Inject Stop** → 停止
 - ☐ **Set Adapter Mode: Step / Follow** → SmfPlayerClockAdapter 模式切换
 - ☐ onBeat 事件出现在日志中
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8. ScalePracticeSampleScene

路径: Assets/MIDI/Samples/Gameplay/Scenes/ScalePracticeSampleScene.unity

目的: 全部音阶类型的从属判定

- ☐ 切换音阶类型 (Major / Minor 等) → In scale 显示变化
 - ☐ **Add C4 (60)** → In scale: Yes (选择 Major 时)
 - ☐ **Add C#4 (61)** → In scale: No, Scale violation 日志
 - ☐ **Add E4 (64)** → 和弦识别显示更新
 - ☐ **Release All** 清除状态
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9. InputSystemBridgeSampleScene

路径: Assets/MIDI/Samples/Integrations/InputSystem/Scenes/InputSystemBridgeSampleScene.unity

前提: FEATURE_INPUT_SYSTEM

- ☐ Play 开始后生成 Synthetic Device
 - ☐ MIDI Inject 反映到 Input System 状态
 - ☐ Input Action → MIDI 发送 (双向桥接)
 - ☐ 在 Monitor 中确认 IN/OUT
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10. MidiNetworkJamSampleScene

路径: Assets/MIDI/Samples/Integrations/Networking/Scenes/MidiNetworkJamSampleScene.unity

前提: FEATURE_MIDI_NETWORK

- ☐ 通过 Hub / Client 环回或两个实例连接
 - ☐ Broadcast 模式下 MIDI 事件注入到 Client 侧虚拟设备
 - ☐ 切换 Merge / Playback 模式 (若有 UI)
 - ☐ 在 Monitor 中确认 Client 侧 IN
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11. FoundationSampleScene

路径: Assets/MIDI/Samples/Foundation/Scenes/FoundationSampleScene.unity

目的: 设备选择、延迟校准、Foundation UI

- ☐ Play 开始后显示 Foundation UI shell
- ☐ 设备选择 UI 列出已连接设备
- ☐ 延迟校准流程可工作
- ☐ 设置持久化 (若有) 可工作

相关文档

- [示例项目](#)
- [类型套件](#)
- [Unity 生态系统集成](#)
- [面向游戏玩法的组件](#)